Hodge Deligne Numbers for Hypersurfaces in Toric Varieties

Will Gilroy

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1 Mike Meeting 03 Aug

Mike took us through some of the basic ideas. It was a long meeting, but there were a lot of ideas, and so this is a writeup for some of the ideas from that meeting.

Any text in purple indicates either a question from myself or an incomplete thought.

1.1 Definition of Hodge-Deligne Numbers and Some Basic Properties

Let X be a variety over $\mathbb{C}$ of dimension[1] n. The *Hodge-Deligne numbers* denoted $e^{p,q}(X)$ for $0 \leq p, q \leq n$ are a collection of integers satisfying properties soon to be described. Moreover the polynomial $e_X(x, y) := \sum_{0 \leq p,q \leq n} e^{p,q}(X)x^p y^q$ is called the *Hodge-Deligne* polynomial. The Hodge-Deligne numbers and polynomial satisfy the following properties

- The Hodge-Deligne polynomial is symmetric in x, y, $e_X(x, y) = e_X(y, x)$. In other words, the Hodge-Deligne numbers are symmetric in p, q, $e^{p,q}(X) = e^{q,p}$.
- If X is reducible then the Hodge-Deligne polynomial decomposes additively over the components: if $X = Y \amalg Z$ (presumably these have to be closed subsets?) then $e_X = e_Y + e_Z$. (A variety is by definition irreducible, so, perhaps the most general setting is for a separated, finite type over $\mathbb{C}$-scheme? Or maybe this property is supposed to be for closed subvarieties?)
- If X is a product variety then its Hodge-Deligne polynomial decomposes multiplicatively: if $X = Y \times_{\mathbb{C}} Z$ then $e_X = e_Y e_Z$.
- If X is projective and smooth (over $\mathbb{C}$) then recall that the hodge numbers are the dimensions of the $\mathbb{C}$-vector spaces given by cohomology over the sheaf of differentials

$$h^{p,q}(X) = \dim_{\mathbb{C}} H^q(X, \Omega^p_X).$$

[1] A variety is defined to be separated, irreducible, finite type k-scheme. I am trying to remember why we have that a variety has a well-defined dimension.

In this setting, the Hodge-Deligne numbers are related to the Hodge numbers in the following way

$$e^{p,q}(X) = (-1)^{p+q} \cdot h^{p,q}(X).$$

1.2 Examples

Example 1 (A point)**.** Let $X = \{*\} = \mathrm{Spec}(\mathbb{C})$. We have that X is smooth and projective. The following Hodge calculation will give that $e_X(x, y) = 1$.

Since X is affine $H^q(X, \Omega_X^p)$ is non-vanishing only for $q = 0$. Recall that $\Omega_X^p = 0$ for all $p > 0$ (I have forgotten why this is true, but it follows because X is a point. I think it should be easy to build the module of differentials in this case once one remembers the definition and construction) and so $h^{p,q} = 0$ for all $p, q > 0$. The only possible non-zero Hodge number we may have is for $p = q = 0$. We have the following

$$\begin{aligned}
h^{0,0}(X) &:= H^0(X, \Omega_X^0) \\
&= \dim_{\mathbb{C}} H^0(X, \mathcal{O}_X) && \text{By definition} \\
&= \dim_{\mathbb{C}} \Gamma(X, \mathcal{O}_X) && \text{By the cohomology axioms} \\
&= \dim_{\mathbb{C}} \mathbb{C} && X = \mathrm{Spec}(\mathbb{C}) \\
&= 1.
\end{aligned}$$

Overall, we have a that $h^{0,0}(X) = 1$ and $h^{p,q}(X) = 0$ for all other p, q. The claimed result follows.

Example 2 (Union of Points)**.** It follows from the previous example and the decomposition property of e_X that a disjoint union of n points has Hodge-Deligne polynomial $e_X(x, y) = n$.

Example 3 ($P^1_{\mathbb{C}}$)**.** Performing a similar Hodge computation to the example of the point one can compute that

$$e_{\mathbb{P}^1_{\mathbb{C}}} = 1 + xy.$$

This might eb a worthwhile computation. $\mathbb{P}^1$ is covered by two affine patches and so we only need to compute for $p < 2$. We need to recall how to compute the sheaf of differentials however, but I think it should be nonvanishing only degree $0, 1$.

Example 4 ($\mathbb{A}^1_{\mathbb{C}}$)**.** We can decompose $P^1 = \mathbb{A}^1 \amalg \{*\}$ the affine line with a "point at infinity". "point at infinity." Since $\mathbb{A}^1$ is an open subset and $\{*\}$ is a closed point we have

$$e_{\mathbb{A}^1_{\mathbb{C}}}(x, y) = e_{\mathbb{P}^1_{\mathbb{C}}} - e_{\{*\}} = xy.$$

~~I'm a bit confused by this example, becuase the affine line is affine, and so we should expect it to only~~

~~have have cohomology in degree 0. That is to say, I would expect that this should be a polynomial only in y.~~ The Hodge-Deligne numbers agree with the Hodge numbers only when X is projective – $\mathbb{A}^1$ is not projective, so we're gucci.

Example 5 ($\mathbb{A}^n_{\mathbb{C}}$.)**.** Using the product property of the Hodge-Deligne polynomial we have

$$E_{\mathbb{A}^n} = (xy)^n.$$

From this example, we can read off the hodge numbers without needing to do any Hodge comptuations.

Example 6 (Algebraic Torii)**.** We can decompose the affine line $\mathbb{A}^1 = \mathbb{C}^* \amalg \{*\}$ and so using the decomposition property of the Hodge-Deligne polynomial we have

$$e_{\mathbb{C}^*} = e_{\mathbb{A}^1_{\mathbb{C}}} + e_{\{*\}} = xy - 1.$$

Then, using the product property we further have that

$$e_{(\mathbb{C}^*)^n} = (e_{\mathbb{C}^*})^n = (xy-1)^n.$$

Example 7 ($\mathbb{P}^n_{\mathbb{C}}$)**.** Recall that we can decompose $\mathbb{P}^n = \mathbb{C}^n \amalg \mathbb{P}^{n-1}$ and then we can find the Hodge-Deligne polynomial of $\mathbb{P}^n$ inductively. Consider,

$$\begin{aligned} e_{\mathbb{P}^1} &= 1 + xy \\ e_{\mathbb{P}^2} &= e_{\mathbb{C}^2} + e_{\mathbb{P}^1} = (xy)^2 + (1+xy) \\ e_{\mathbb{P}^3} &= e_{\mathbb{C}^3} + e_{\mathbb{P}^2} = (xy)^3 + (1 + xy + (xy)^2) \\ &\dots \\ e_{\mathbb{P}^n} &= 1 + (xy) + (xy)^2 + \cdots + (xy)^n. \end{aligned}$$

2 Procedure for Hypersurfaces in Toric Varities.

we should go back and fill out some of the preamble thoery here

Suppose we have a full dimension polytope Δ. We can decompose our toric variety

$$P_\Delta = \amalg_{\Gamma' \leq \Delta} T_{\Gamma'}$$

where $T_{\Gamma'} = (\mathbb{C}^*)^{\dim \Gamma'}$. And so we can compute the Hodge-Deligne polynomial of a toric variety using the decomposition property. Does each subcone really just give a torus in a toric variety? Or is it somehow sufficient to compute the Hodge-Deligne polynomial on a dense open?

Let $Z_{\Gamma'} = Z \cap T_{\Gamma'}$. We have the following formula

$$e^{pq}(\bar{Z}) = e^{pq}(Z) + \sum_{\Gamma' < \Delta} e^{pq}(Z_{\Gamma'}).$$

Note that a consequence of this formula is that for $p = n$ or $q = n$, $e^{pq}(\bar{Z})$ is determined by $e^{pq}(Z)$. I think this should follow from the dimension property of Hodge-Deligne numberes, and each $e^{pq}(Z_{\Gamma'})$ has dimension strictly less than n. Is this reasoning correct?

We also have the following formulae

$$e^{pq}(Z) = e^{p+1,q+1}(T) = \begin{cases} 0 & p \neq q \\ (-1)^{n+p+1}\binom{n}{p+1} & p = q \end{cases}$$

And for $p + q \geq n$,

$$\sum_{q=0}^{n-1} e^{pq}(Z) = (-1)^p \left((-1)^{n+1} \binom{n}{p+1} - \sum_{k \geq 1} (-1)^k \binom{n+1}{p+k+1} \ell^*(k\Delta) \right). \tag{*}$$

where ℓ^* denotes the number of interior lattice points of the given polytope. Is this last formula actually correct?

I think the algorithm is then as follows: we iteratively compute the Hodge-Deligne numbers starting from the smalles cone, and we only need to use formula $(*)$ to add the numbers in degree n, is that correct?